

Software USB communication controller for UPS PowerCom IMPERIAL series

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Table 1. Description of the HID Reports for PowerCom UPS-USB interface

Report number	Page, Usage ID	Usage Name	Data Access
	8404	UPS COLLECTION (Application)	
	8424	PowerSummary COLLECTION (Logical)	
1	84FE	iProduct	R/O
2	84FF	iSerialNumber	R/O
3	84FD	iManufacturer	R/O
4	858F	iOEMInformation	R/O
5	8589	iDeviceChemistry	R/O
7	858B	Rechargeable	R/O
8	852C	CapacityMode	R/O
9	8583	DesignCapacity *	R/O
	8567	FullChargeCapacity *	
10	8566	RemainingCapacity	R/O
11	858C	WarningCapacityLimit	R/O
12	8529	RemainingCapacityLimit	R/O
13	8585	ManufacturerDate	R/O
14	8568	RunTimeToEmpty	R/O
15	8457	DelayBeforeShutdown	R/W
16	8456	DelayBeforeStartup	R/W
17	8440	ConfigVoltage	R/O
18	8430	Voltage	R/O
19	845A	AudibleAlarmControl	R/O
20	8402	PresentStatus COLLECTION (Logical)	R/O
		Collection multi-bits Report **	
	8544	Charging	
	8545	Discharging	
	85D0	ACPresent	
	85D1	BatteryPresent	
	8542	BelowRemainingCapacityLimit **	
	8543	RemainingTimeLimitExpired **	
	854B	NeedReplacement	
	85DB	VoltageNotRegulated **	
	8468	ShutdownRequested **	
	8469	ShutdownImminent	
	8473	CommunicationLost	
	8465	Overload	
		COUNT (3) **	
	8680	Usage (80) from Usage Page (86) **	
		END_COLLECTION [PresentStatus, Logical]	
		END_COLLECTION [PowerSummary, Logical]	
	8412	Battery COLLECTION (Physical)	
21	8458	Test	R/W
22	8470	Initialized	W/O
23	8585	ManufacturerDate	R/O
24	8566	RemainingCapacity	R/O
25	8440	ConfigVoltage	R/O
26	8430	Voltage	R/O
27	8456	DelayBeforeStartup	R/W
		END_COLLECTION [Battery, Physical]	

Report number	Page, Usage ID	Usage Name	Data Access
	841A	Input COLLECTION (Physical)	
28	8440	ConfigVoltage	R/O
29	8430	Voltage	R/O
30	8432	Frequency	R/O
		END_COLLECTION [Input, Physical]	
	841C	Output COLLECTION (Physical)	
31	8435	PercentLoad	R/O
32	8440	ConfigVoltage	R/O
33	8430	Voltage	R/O
34	8432	Frequency	R/O
35	8457	DelayBeforeShutdown	R/W
36	8456	DelayBeforeStartup	R/W
		END_COLLECTION [Output, Physical]	
6	8401	iName	R/O
37	845A	Audible Alarm Control	R/O
38	8402	Present Status COLLECTION (Logical)	R/O
		Collection multi-bits Report	
	8544	Charging	
	8545	Discharging	
	85D0	ACPresent	
	85D1	BatteryPresent	
	8542	BelowRemainingCapacityLimit **	
	8543	RemainingTimeLimitExpired **	
	854B	NeedReplacement	
	85DB	VoltageNotRegulated **	
	8468	ShutdownRequested **	
	8469	ShutdownImminent	
	8473	CommunicationLost	
	8465	Overload	
	846E	Boost	
	846F	Buck	
	8471	Tested	
	8680	Usage(80) from Usage Page (86) **	
		END_COLLECTION [PresentStatus, Logical]	
39	8469	ShutdownImminent	R/W
		END_COLLECTION [UPS, Application]	

*) - two Usage in one report.

***) - Not to use. For compatibility with system driver.

Table 1.a. Not standard Usages.

String number	Usage Name	Data Access
100	Usage ReadUART	R/O

Table 2. Data for Usage from UART of Power Devices PowerCom and constants.

System requests for usage with Microsoft UPS system driver
(first Byte is Byte 0)

Report number	Page, Usage ID	Usage Name	Data Access	Data	Length
	8424	PowerSummary COLLECTION			
1	84FE	iProduct	R/O	UPS IMP-XXXXAP FW1.A0 XXXX - Model number from table 3	60 Byte
2	84FF	iSerialNumber	R/O	0A2-0000-0001	32 Byte
3	84FD	iManufacturer	R/O	POWERCOM Co.,LTD	34 Byte
4	858F	iOEMInformation	R/O	POWERCOM Co.,LTD	34 Byte
5	8589	iDeviceChemistry	R/O	PbAc	10 Byte
7	858B	Rechargable	R/O	1	1 Byte
8	852C	CapacityMode	R/O	2 (%)	1 Byte
9	8583	Designcapacity*	R/O	100 (in %)	2 Byte
	8567	FullChargeCapacity *		100 (in %)	
10	8566	RemainingCapacity	R/O	Byte 1 "battery level" (in %)	1 Byte
11	858C	WarningCapacityLimit	R/O	30 (in %)	1 Byte
12	8529	RemainingCapacityLimit	R/O	Initially 10 (in %), is it hereinafter assigned by system driver	1 Byte
13	8585	ManufacturerDate	R/O	(Year-1980) * 512 + Month * 32 + Day	2 Byte
14	8568	RunTimeToEmpty	R/O	It Is Calculated on the grounds of Byte 0, Byte 1, Byte 12, (in sec) Calculated in table $T = ((Caccum * K * BatteryLevel * 60 \text{ min}) / (Pnom * LoadLevel)) * 60$ Where K =9,6 for Imp425-Imp825 K =19,2 for Imp1025-Imp2000 Unit Caccum – 4%, Unit LoadLevel – 16%,	2 Byte

Report number	Page, Usage ID	Usage Name	Data Access	Data	Length
15	8457	DelayBeforeShutdown ***	R/W	Read: Byte 13, Byte 14 (min, sec) Write: <i>Two free senior bits of byte of seconds are used for possibility of a choice of a prefix-command before 188 command performance</i> If Byte(sec), bit7=0 and bit6=0 Then If Byte 9, bit0=1 Then command 185, 188, min, sec If Byte 9, bit0=0 Then command 186, 188, min, sec If Byte(sec), bit7=0 and bit6=1 Then command 185, 188, min, sec If Byte(sec), bit7=1 and bit6=0 Then Then command 186, 188, min, sec If Byte(sec), bit7=1 and bit6=1 Then No actions	2 Byte
16	8456	DelayBeforeStartup ***	R/W	Read: Byte 7, byte 8 (min) Write: Command 4, high byte min, low byte min	2 Byte
17	8440	ConfigVoltage	R/O	from Byte 12 (see table 3)	1 Byte
18	8430	Voltage	R/O	Byte 3 " output Voltage "	2 Byte
19	845A	AudibleAlarmControl	R/O	If Byte 10, bit3=1 Then value =2 If Byte 10, bit3=0 Then value =1	1 Byte
	8412	Battery COLLECTION			
21	8458	Test	R/W	Read: If Byte 10 bit1=0 AND bit2=0 Then value = 1 ("Done and Passed") If Byte 10 bit1 =1 AND bit2=0 Then value = 3 ("Done and Error") If Byte 10 bit2 =1 Then value = 5 ("Test in progress") Write : If value =1 Then command 3 , If value<>1 Then "No test"	1 Byte

Report number	Page, Usage ID	Usage Name	Data Access	Data	Length
22	8470	Initialized	W/O	Write : If value =1 Then command 11 , Else Not command	1 Byte
23	8585	ManufacturerDate	R/O	(Year-1980) * 512 + Month * 32 + Day	2 Byte
24	8566	RemainingCapacity	R/O	Byte 1 " battery level " (in %)	1 Byte
25	8440	ConfigVoltage	R/O	from Byte 12 (see table 3) (in Volt)	1 Byte
26	8430	Voltage	R/O	Byte 3 "output Voltage" (in Volt)	2 Byte
27	8456	DelayBeforeStartup ***	R/W	Similary Report number 16	2 Byte
	841A	Input COLLECTION			
28	8440	ConfigVoltage	R/O	from Byte 12 (see table 3) (in Volt)	1 Byte
29	8430	Voltage	R/O	Byte 2 "input Voltage" (in Volt)	2 Byte
30	8432	Frequency	R/O	Byte 4 "input Frequency" (in Hz)	1 Byte
	841C	Output COLLECTION			
31	8435	PercentLoad	R/O	Byte 0 "load level" (in %)	1 Byte
32	8440	ConfigVoltage	R/O	from Byte 12 (see table 3) (in Volt)	1 Byte
33	8430	Voltage	R/O	Byte 3 "output Voltage" (in Volt)	2 Byte
34	8432	Frequency	R/O	Byte 6 "output Frequency" (in Hz)	1 Byte
35	8457	DelayBeforeShutdown ***	R/W	similarly Report number 15	2 Byte
36	8456	DelayBeforeStartup ***	R/W	similarly Report number 16	2 Byte
6	8401	iName	R/O	POWERCOM USB UPS	34 Byte
37	845A	Audible Alarm Control	R/O	If Byte 10, bit3=1 Then value =2 If Byte 10, bit3=0 Then value =1	1 Byte
	8402	Present Status COLLECTION			
38		Collection multi-bits Report	R/O		2 Byte
	8544	Charging		(see table 4)	
	8545	Discharging		(see table 4)	
	85D0	ACPresent		Byte 9, Not bit0	
	85D1	BatteryPresent		(see table 4)	
	8542	BelowRemainingCapacityLimit **		(see table 4)	

Report number	Page, Usage ID	Usage Name	Data Access	Data	Length
	8543	RemainingTimeLimitExpired **		(see table 4)	
	854B	NeedReplacement		(see table 4)	
	85DB	VoltageNotRegulated		Byte 10, bit0	
	8468	ShutdownRequested **		0	
	8469	ShutdownImminent		IF Byte 10, bit6=0 and [Byte 0]=0 Then value =1	
	8473	CommunicationLost		IF Error UART Then value =1	
	8465	Overload		Byte 9, bit5	
	846E	Boost		Byte 9, If bit3=1 then value=bit4, If bit3=0 Then value=0	
	846F	Buck		Byte 9, If bit3=1 then value= Not bit4, If bit3=0 Then value=0	
	8471	Tested		Byte 10, bit2	
39	8469	ShutdownImminent	R/W	Read: IF [Byte 10, bit6=0] and [Byte 9, bit0=1] and [Byte 0]=0 Then value =1 In all other cases value = 0 Write: If value=1 Then command "G" Else command "g"	1 Byte

String number	Usage Name	Data Access	Data	Length
100	Usage ReadUART	R/O	Read data from UART (Byte 0 ÷ Byte 15)	16 Byte

*) - two Usage in one report.

**) - Not to use. For compatibility with system driver.

***) - Does not correspond to the standard "Universal Serial Bus Usage Tables for HID Power Devices". Productivity microcontroller CY7C63723 is not sufficient, that real-time to execute transformations data.

****) - string variable are sent in format UNICOD

Table 3. UART. Byte 12. UPS model.

higher part of the byte - UPS model, Pnom and Design Capacity,

higher part of the byte	Model	Model number	Pnom (Wt)	DesignCapacity (A/H)	Number Acc.
1	Imp-425	-425	425	7	1
2	Imp-525	-525	525	7	1
3	Imp-625	-625	625	7	1
4	Imp-825	-825	825	9	1
5					
6	Imp-1025	1025	1025	7	2
7	Imp-1200	1200	1200	7	2
8	Imp-1500	1500	1500	7	2
A	Imp-2000	2000	2000	9	2

lower part of the byte - Config Voltage.

lower part of the byte	ConfigVoltage (Volt)
0	100
1	110
2	115
3	120
8	220
9	230
A	240

Table 4. Conditions for setting Present Status.

		HEX	USAGE (80)	Tested	Buck	Boost	Overload	Communication Lost	Shutdown Imminent	Shutdown Requested	Voltage Not Regulated	Need Replacement	Remaining Time Limit Expired	Below Remaining Capacity Limit	Battery Present	AC Present	Discharging	Charging
			7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
1.	<u>Fully Charged</u> Byte 9, bit0 = 0 BatteryLevel = 100%	XX 0C	T*	T*	T*	T*	T*	T*	T*	T*	T*	0	0	0	1	T*	0	0
2.	<u>Discharging</u> Byte 9, bit0 = 1 10% ≤ BatteryLevel ≤ 100%	XX 0A	T*	T*	T*	T*	T*	T*	T*	T*	T*	0	0	0	1	T*	1	0
3.	<u>Discharging</u> Byte 9, bit0 = 1 0% < BatteryLevel ≤ 10%	XX 2A	T*	T*	T*	T*	T*	T*	T*	T*	T*	0	1	0	1	T*	1	0
4.	<u>Fully Discharged</u> Byte 9, bit0 = 1; Byte 9, bit1 = 0 BatteryLevel = 0%	XX 3A	T*	T*	T*	T*	T*	T*	T*	T*	T*	0	1	1	1	T*	1	0
5.	<u>Fully Discharged</u> Byte 9, bit0 = 1; Byte 9, bit1 = 1 BatteryLevel = 0%	XX 78	T*	T*	T*	T*	T*	T*	T*	T*	T*	1	1	1	1	T*	0	0
6.	<u>Charging</u> Byte 9, bit0 = 0; Byte 9, bit1 = 1 BatteryLevel = 0%	XX 7D	T*	T*	T*	T*	T*	T*	T*	T*	T*	1	1	1	1	T*	0	1
7.	<u>Charging</u> Byte 9, bit0 = 0; Byte 9, bit1 = 0 BatteryLevel = 0%	XX 3D	T*	T*	T*	T*	T*	T*	T*	T*	T*	0	1	1	1	T*	0	1
8.	<u>Charging</u> Byte 9, bit0 = 0 0% < BatteryLevel ≤ 7%	XX 2D	T*	T*	T*	T*	T*	T*	T*	T*	T*	0	1	0	1	T*	0	1
9.	<u>Charging</u> Byte 9, bit0 = 0 7% < BatteryLevel ≤ 99%	XX 0D	T*	T*	T*	T*	T*	T*	T*	T*	T*	0	0	0	1	T*	0	1

T*) – from Table 2. Data for Usage from UART of Power Devices PowerCom and constants.

Position of SN (serial number) in file of firmware (PCM.hex or PCM.rom) – OB13.

Fragment of listing below:

```
0B11      iserialnumber_string:      ; Serial Number String
0B11 20 [00] db 20h
0B12 03 [00] db 3
0B13 30 00 ... dsu "0A2-0000-0001 "
```